NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 22/03/2026
Time: 3 Hrs | GTM – 01 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	B	B	B	B	A	C	A	C	A
11	12	13	14	15	16	17	18	19	20
D	B	A	B	A	D	A	B	C	B
21	22	23	24	25					
3	2	25	83	1					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	B	C	C	C	B	A	C	A	D
36	37	38	39	40	41	42	43	44	45
D	C	D	C	A	D	B	B	B	D
46	47	48	49	50					
5	925	78	75	146					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	D	B	C	A	B	D	A	B	C
61	62	63	64	65	66	67	68	69	70
D	D	C	C	A	D	B	D	D	C
71	72	73	74	75					
3	3	1	1	1					

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HINTS
PHYSICS

1. $F = ax^2 = bt^{\frac{1}{2}}$

$$[a] = \frac{[F]}{[x^2]} = \left[M^1L^{-1}T^{-2}\right]$$

$$[b] = \frac{[F]}{\left[t^{\frac{1}{2}}\right]} = \left[M^1L^{-1}T^{-\frac{5}{2}}\right]$$

$$\left[\frac{b^2}{a}\right] = \frac{\left[M^2L^2T^{-5}\right]}{\left[M^1L^{-1}T^{-2}\right]} = \left[M^1L^3T^{-3}\right]$$

2. Average value over the distance 0 to d:

$$\bar{u} = \frac{1}{d}\int_0^d u(x)dx$$

$$\bar{u} = \frac{1}{2}\varepsilon_0\frac{V^2}{d^2}\cdot\frac{1}{d}\int_0^d\left(1+\frac{x}{d}\right)dx$$

$$\int_0^d\left(1+\frac{x}{d}\right)dx = d+\frac{d}{2} = \frac{3d}{2}$$

$$\bar{u} = \frac{1}{2}\varepsilon_0\frac{V^2}{d^2}\cdot\frac{3}{2}$$

3. Initial power $P = 4.0\,D$

Initial focal length $f = 1/P = 1/4 = 0.25\,m$

New power $P' = 4.0 + 0.2 = 4.2\,D$

New focal length $f' = 1/4.2 \approx 0.2381m$

Relative decrease in focal length

$= (f - f')/f = (0.25 - 0.2381)/0.25 \approx 0.0119/0.25 \approx 0.0476 \approx 0.05$

So the correct answer is (2)0.05 (approximately 5% decrease).

4. Average K.E of each molecule is $E_{av} = E\frac{3}{2}K_BT$

If N_1 is the number of molecules of hydrogen and N_2 is number of molecules of oxygen, then Total K.E of hydrogen $E_1 = N_1\left(\frac{3}{2}K_BT\right)$

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Total K.E of oxygen is $E_2 = N_2\left(\frac{3}{2}K_BT\right)$

$$\Rightarrow \frac{E_1}{E_2} = \frac{N_1}{N_2} = 2$$

5. $\frac{1}{f_1} = (1.5-1)\left[\frac{1}{2}-0\right] \Rightarrow f_1 = 4cm$

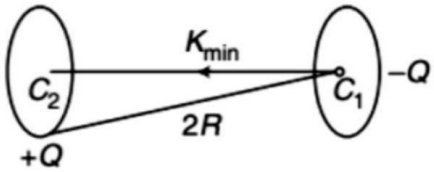
$$\frac{1}{f_2} = \left(\frac{1.5}{1.2}-1\right)\left(\frac{1}{3}-0\right)$$

$$\frac{1}{f_2} = \frac{0.3}{1.2}\times\frac{1}{3}$$

$$f_2 = 12$$

$$f_1 : f_2 = 4:12 = 1:3$$

6.



$$k_{C_1} + U_{C_1} = k_{C_2} + U_{C_2}$$

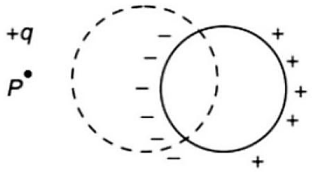
$$k_{min} + q\,V_{C_1} = 0 + q\,V_{C_2}$$

$$V_{C_1} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{2R}-\frac{Q}{R}\right)$$

$$V_{C_2} = \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{R}-\frac{Q}{2R}\right)$$

Substituting these values in Eq. (i), we can find K_{min}

7.



The induced charges on conducting sphere due to $+q$ charge at P are as shown in figure. Now, net charge inside the closed dotted surface is negative. Hence, according to Gauss's theorem net flux is negative.

8. $p^2V =$ constant

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$\therefore \left(\frac{nRT}{V}\right)^2 V = \text{constant}$

$\therefore T^2 \propto V$

or $T \propto \sqrt{V}$

V is made three times. So, T will becomes $\sqrt{3}$ times.

9. For the sector, $y_{cm} = \frac{4R}{3}\frac{\sin(\phi/2)}{\phi} = \frac{2R}{\pi}\left(\because \phi = \frac{\pi}{3}\right)$

For the arc, $y_{cm} = \frac{2R\sin(\phi/2)}{\phi} = \frac{3R}{\pi}$

10. $V = \frac{V_0 t}{T/4} = \frac{4V_0 t}{T}$

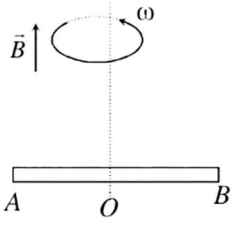
$$V_{rms} = \sqrt{\langle V^2 \rangle} = \frac{4V_0}{T}\sqrt{\frac{\int_0^{T/4} t^2 dt}{\int_0^{T/4} dt}} = \frac{V_0}{\sqrt{3}}$$

11. Potential difference between

O and A is $V_A - V_O = \frac{1}{2}Bl^2\omega$

O and B is $V_B - V_O = \frac{1}{2}Bl^2\omega$

So $V_A - V_B = 0$



12. $\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$

$\Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} \Rightarrow \frac{10^{-10}}{0.5\times10^{-10}} = \sqrt{\frac{E_2}{E_1}}$

$\Rightarrow E_2 = 4E_1$

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Hence added energy $= E_2 - E_1 = 3E_1$

13. Using Moseley's law, we get

$$\frac{\lambda_2}{\lambda_1} = \frac{(Z_1 - a)^2}{(Z_2 - a)^2}$$

$$\lambda_2 = \frac{200 \times (74-1)^2}{(78-1)^2} = 179.76\,\text{Å}$$

14. From snell's law

$$\mu \sin\theta = 1.6\mu \sin x$$

$$\Rightarrow \sin X = \frac{5}{8}\sin\theta$$

15. $Q = A_{He} \cdot BE_{He}$ (per nucleon) $-2A_H (BE_H)$ (per nucleon)

= 28 MeV – 4.4 MeV = 23.6 MeV

16. Collective energy of both photons is 11.6 eV more than the ionization energy of a hydrogen atom.

17. Momentum of two electrons are $\frac{h}{\lambda_1}\hat{i}$ and $\frac{h}{\lambda_1}\hat{j}$

Velocity of centre of mass $\vec{V}_{CM} = \frac{h}{2m\lambda_1}\hat{i} + \frac{h}{2m\lambda_2}\hat{j}$

Velocity of first electron about centre of mass is

$$\vec{V}_{CM} = \frac{h}{2m\lambda_1}\hat{i} - \frac{h}{2m\lambda_2}\hat{j}$$

$$\Rightarrow \lambda_{CM} = \frac{h}{\sqrt{\frac{h^2}{4\lambda_1^2} + \frac{h^2}{4\lambda_2^2}}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

18. . Let m be the mass of the disc. Then translational kinetic energy of the disc is

$$K_T = \frac{1}{2}mv^2 .....(1)$$

As it ascends on a smooth track its rotational kinetic energy will remain same while translational

kinetic energy will go on decreasing. At highest point.

$$K_T = mgh$$

$$\Rightarrow \frac{1}{2}mv^2 = mgh$$

$$\Rightarrow h = \frac{v^2}{2g} = \frac{(6)^6}{2\times 10} = 1.8m$$

19. $P_{max} = V_z \times I_{z(max)}$ 20. $\frac{4}{3}\pi R^3 = 10\left(\frac{4}{3}\pi r^3\right)$

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$\Rightarrow R = (10^2 r)$

Now, $U_i = 10^6 (4\pi r^2)T$ and $U_f = (4\pi r^2)T$

$\Rightarrow \dfrac{U_f}{U_i} = \dfrac{1}{10^2}$

21. Pseudo force and friction will make a force couple. Normal reaction and Mg will make a force couple.

When block is about the topple normal reaction will shift to edge.

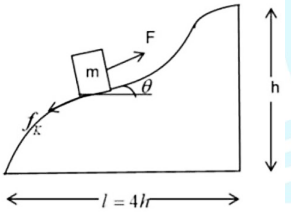
22. $\Delta W_f = -\int \mu mg \cos\theta ds$

$= -\mu mg \int_0^l dx$

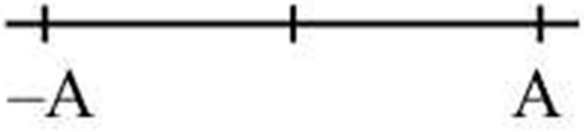
$= -\mu mgl = -0.5mg(4h) = -2mgh$

$\Delta W_g = -mgh$

$\therefore \Delta W_{fr} / \Delta W_g = 2$



23. $A = 1cm$



$n = \dfrac{12.5}{2} = 6.25$ cycles

$\therefore D = 4 \times 6 + 1 = 25$

$d = 1$

$\dfrac{D}{d} = 25$

24. $W = \int_{\vec{r}}^{\vec{r}} \vec{F}.d\vec{r}, where\, d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$

$W = \int_{\vec{r}_1}^{\vec{r}_2} \left(3x^2\hat{i} + 2y\hat{j}\right).\left(\hat{i}dx + \hat{j}dy + \hat{k}dz\right)$

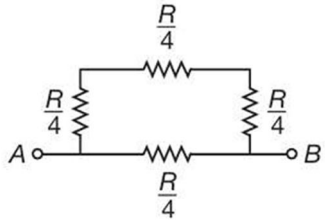
$W = \int_{\vec{r}_1}^{\vec{r}_2} \left(3x^2 dx + 2ydy\right)$

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$$W=\int_{(2,3)}^{(4,6)} d\left(x^3\right)+\int_{(2,3)}^{(4,6)} d\left(y^2\right)=\left(x^3+y^2\right)\Big|_{(2,3)}^{(4,6)}$$

$$W=83J$$

25.



$$\Rightarrow R_{AB}=\frac{3R}{16}=\frac{3(16)}{16}=3\Omega$$

CHEMISTRY

26. $\underset{0.3\text{ mole}}{Zn(s)}+\underset{0.52\,\text{mole(limiting)}}{2HCl(aq.)}\longrightarrow ZnCl_2(aq.)+\underset{0.26\text{ mole}}{H_2(g)}$

27. $\dfrac{E_2-E_1}{E_4-E_3}=\dfrac{\frac{1}{1^2}-\frac{1}{2^2}}{\frac{1}{3^2}-\frac{1}{4^2}}=15$ approx

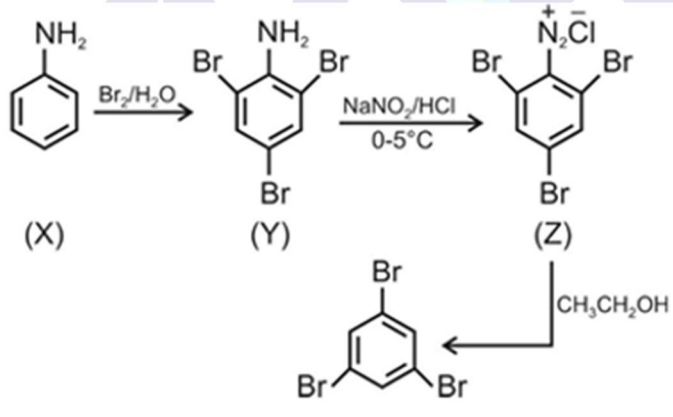
28. (A) Bond order is directly proportional to the bond energy.

Bond order of $N_2=3, N_2^+, N_2^-=2.5$ $N_2^{2-}=2$

But N_2^- has more electrons in antibonding MO's and thus N_2^+ is more stable than N_2^-. So correct order of bond energy will be $N_2>N_2^+>N_2^-$

29. Electron withdrawing groups destabilize the carbocation.

30.



31. $6\times10^{-4}\times1=\dfrac{4}{m}\times0.0821\times300\Rightarrow m=1.6\times10^5$

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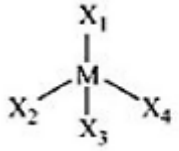
32. $K_{eq} = \frac{K_1}{K_2} = \frac{(b+x)}{(a-x)} \Rightarrow x = \frac{k_1 a - k_2 b}{k_1 + k_2}$

33. HI has donated H^+ in the above reaction. So it acts as an Arrhenius acid as well as a Bronsted acid.

34. $\frac{10}{90} = \frac{W_{Na}}{10} \Rightarrow W_{Na} = \frac{10}{9}g$

$\frac{10/9}{23} = \frac{10 \times t}{96500} \Rightarrow t = 466.18\ s = 7.77\ min$

35.



Bond angles

Total angles are six [X_1MX_2, X_1MX_3, X_1MX_4, X_2MX_3, X_2MX_4, X_3MX_4]

36. Eq. of NaH_2PO_3 + Eq. of $NaHCO_3$ = Eq. Of NaOH

$\frac{20 \times 0.1}{1000} \times 1 + \frac{40 \times 0.1}{1000} \times 1 = x \Rightarrow x = 6 \times 10^{-3}$

37. As there are electrons in the valence shell of group 13 elements they show + 3 oxidation state. Also in case of heavier members, due to inert pair effect, +1 oxidation state is most stable.

38. The reaction of concentrated H_2SO_4 on all the compounds is as follows.

$HCOOH \xrightarrow[H_2SO_4]{conc.} H_2O + CO$

$H_2C_2O_4 \xrightarrow[H_2SO_4]{conc.} CO_2 + CO + H_2O$

In these reactions, concentrated H_2SO_4 acts as a dehydrating agent.

$K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \rightarrow 2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$

39. II and III are C-4 epimer

40. (A) Coordination number of Pt is 6, hence

I $K_2[PtCl_6]$ - Three ions

II $[Pt(NH_3)_2Cl_4]$ - Zero ions

III $[Pt(NH_3)_3Cl_3]Cl$ - Two ions

IV $[Pt(NH_3)_5Cl]Cl_3$ - Four ions

Conductivity $\propto$ no. of ions

41. $C_2H_5Br \xrightarrow{aq\ KOH} C_2H_5OH\ (Product\text{-}I)$

(Nucleophilic substitution reaction)

$C_2H_5Br \xrightarrow{alc\ KOH} C_2H_4\ (Product\text{-}II)$

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(elimination reaction)

C_2H_5OH (Product-I) isomer is CH_3OCH_3

$$C_2H_5OH \xrightarrow{\text{dehydration}} C_2H_4$$

42. $$CH_2 = CH - CH = CH_2 \xrightarrow[\text{1,4 addition}]{H_2} \underset{(A)}{CH_3 - CH = CH - CH_3}$$

$$\xrightarrow{O_3/H_2O} \underset{(B)}{2CH_3COOH}$$

[Zinc is not present hence CH_3CHO first formed is oxidised to CH_3COOH)

43.

$$CH \equiv CH + CH_3MgBr \longrightarrow CH \equiv C - MgBr$$

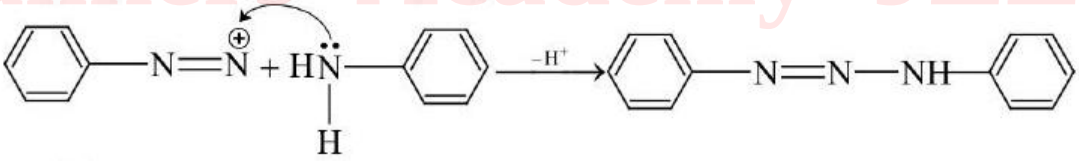
– COOH is (EWG)

$$CH \equiv C \xrightarrow{CO_2/H_3O^+} -COOH$$

$$\xrightarrow{HgSO_4/H_2SO_4} OHC - CH_2 - COOH$$

$$\xrightarrow{Ag_2O} HOOC - CH_2 - COOH$$

44. Diazonium cation reacts with aniline in weakly acidic medium resulting in N, N coupling rather than C-coupling.



45. Markonikoff's rule applicable for unsymmetrical alkenes

46. $(3 \times 300) + (2 \times 4 \times 200) = M(300 + 200) \Rightarrow M = 5\ M$

47. When half of NH_4OH is neutralized, $[NH_4OH] = [NH_4Cl]$. By using the Henderson equation,

$$pOH = pK_b + \log\frac{[Salt]}{[Base]} \text{ or } pOH = 4.75$$

pH = 14 – 4.75 = 9.25

48. Dipole moment of compound would have been completely ionic

$$= \left(4.8 \times 10^{-10}\,\text{esu}\right)\left(2.67 \times 10^{-8}\,\text{cm}\right) = 12.816\text{D}$$

$$\text{so \% ionic character} = \frac{10.0}{12.816} \times 100\% = 78.03\% \simeq 78\%$$

49. At the end of 5 hours

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$A_{t,1} = \frac{A_0}{2}$

at the end of 10th hours

$A_{t,2} = \frac{A_0}{4}$

$A_{t,2} - A_{t,1} = \frac{A_0}{2} - \frac{A_0}{4} = \frac{A_0}{4} = 0.25A_0$

amount decayed = 0.25 mol

moles of O_2 formed = 0.25/2 = 1/8

$W = -P\Delta V = -nRT = -(1/8) \times 300 \times 2Cal = 75Cal$

50. Volume of $\frac{N}{10}HCl$ taken $= 50ml$

Volume of $\frac{N}{10}$ NaOH used for neutralisation of unused acid = 23.2 ml

Now $N_1V_1(NaOH) = N_2V_2(HCl)$

23.2 ml of $\frac{N}{10}NaOH \equiv 23.2ml$ of $\frac{N}{10}HCl$

$\therefore$ Volume of $\frac{N}{10}$ HCl unused = 23.2 ml

$\therefore$ Volume of N/10 HCl required for neutralization of NH_3 = 50 – 23.2 = 26.8 ml

26.8 ml of $\frac{N}{10}$ HCl = 26.8 ml of N/10 NH_3

1000 ml of 1N NH_3 solution contains nitrogen = 14 g

26.8 ml of NH_3 solution contains nitrogen = $\frac{14 \times 26.8}{10 \times 1000}$

Percentage of nitrogen = $\frac{14 \times 26.8 \times 100}{10 \times 1000 \times 0.257}$ = 14.6%

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MATHEMATICS

51. $\frac{1}{a_{n+1}}-\frac{1}{a_n}=n$

$\Rightarrow \frac{1}{a_2}-\frac{1}{a_1}=1, \frac{1}{a_3}-\frac{1}{a_2}=\frac{1}{2}$

$\cdots \frac{1}{a_{1000}}-\frac{1}{a_{999}}=999 \Rightarrow$ adding all

$\frac{1}{a_{1000}}-\frac{1}{a_1}=1+2+\ldots 999$

52.

$f(x)=\frac{(2a-1)x^3+5x^2+3x+(7c-d)}{(a+b)x^4+6x^2+(b-3)x}$

$\Rightarrow f(x)$ is non-zerofinile only

$x \to \infty$ if $2a-1=0$ & $a+b=0$

thus $l_1=\frac{5}{6}$

& f(x) is non zero finite only

$x \to 0$ if $7c-d=0$

thus $l_2=\frac{3}{b-3}=\frac{3}{-7l_2}=-\frac{6}{7}$

$\therefore$ equalion $x^2-\left(\frac{5}{6}-\frac{6}{7}\right)x-\frac{5}{7}=0$

$x^2+\frac{x}{42}-\frac{5}{7}=0$

$42x^2+x-30=0$

53. $f(x)$ is a decreasing function & for major axis to be x-axis

$f(k^2+2k+5) > f(k+11)$

$\Rightarrow k^2+2k+5 < k+11$

$\Rightarrow k \in (-3,2)$

54. Given that $\sum X^2{}_i=400$ and $\sum X_i=80$ if $X_i>0, i=1,2,3,\ldots..,n$

Which are not identical, then using the inequality.

AM of mth power>mth power of AM, if m>1

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$$\Rightarrow \frac{X^2_1 + X^2_2 + \ldots + X^2_n}{n} > \left(\frac{X_1 + X_2 + \ldots + X_n}{n}\right) \Rightarrow \frac{\sum_{i=1}^{n} X^2_i}{n} > \left(\frac{\sum_{i=1}^{n} X^2_i}{n}\right)^2$$

$$\Rightarrow \frac{400}{n} > \left(\frac{80}{n}\right)^2$$

$$\Rightarrow \frac{n^2}{n} > \frac{80^2}{400} \Rightarrow n > 16$$

55. $g(x) = 3 + 4x$

$$g_2(x) = g(g(x) = 3 + 4(3 + 4x)$$

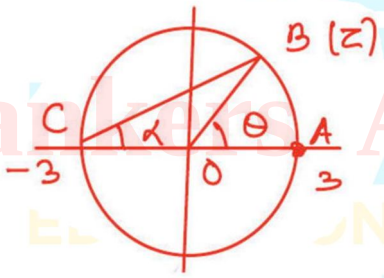
$$= 15 + 4^2 x = 4^2 + 4^2 x - 1$$

$$\Rightarrow g_n(x) = 4^n - 1 + 4^n x = y$$

$$\therefore 4^n x = y - 4^n + 1$$

$$f^{-1}(x) = 4^{-n}\left(x + 1 - 4^n\right)$$

56.



$|\arg z| = \theta$, $|\arg(z+3)| = \alpha$ & by circle prop $\theta = 2\alpha$

57. $S = \{1, 2, \ldots - 9\}$

to get 5 elements, out of 5 odd & 4 even

one odd, $N_1 = {}^5C_1 \cdot {}^4C_4 = 5$

$N_2 = {}^5C_2 \cdot {}^4C_3 = 40$

$N_3 = {}^5C_3 \cdot {}^4C_2 = 60$

$N_4 = 5C_4 \cdot 4C_1 = 20$

$N_5 = 5C_3 = 1$

$\therefore 126$ Numbers

58. $f(x) = \frac{x}{1 + (\ln x)^{\infty}}$, $x \in [1, 3]$

as for $1 \le x < e, 0 \le \ln x < 1\ 4e \le x \le 31 \le \ln x$

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$$\Rightarrow f(x)=\begin{vmatrix} x & 1\le x<e \\ 0 & e\le x\le 3 \end{vmatrix}$$

$\therefore K=e^2$

59.

$(1-\text{Cot}22)(1+\tan 22)(1-\text{Cot}23)(1+\tan 23)$

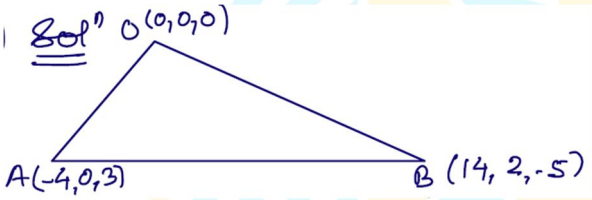
as $(1+\tan A)(1+\tan B)=2$

$(1-\cos A)(1-\cos B)=2$

when $A+B=450$

$\Rightarrow$ Ans is 4

60.



angle bisector through will be along

$$\hat{OA}+\hat{OB}=\frac{(-4i+3k)}{5}+\frac{14i+2j-5k}{15}$$

$$=\frac{2i+2j+4k}{15}$$

$\therefore$ DR must be 1, 1, 2

61. Lel $x\notin y$ be Numbers selected

$\therefore a+b=4\lambda$ & $a-b=4K$

$\Rightarrow a+b$ both are even

$\therefore$ Possibly Paie (0, 4)(0, 8)(4, 8)

(2, 6), (2, 10), (6, 10)

62. $\sum_{r=1}^{n} f(r)=f(1)+f(2)+f(3)+.....+f(n)$

$=f(1)+2f(1)+3f(1)+.....+nf(1)$

Since, $[f(x+y)=f(x)+f(y)]$

$=(1+2+3+.....+n)f(1)$

$=f(1)\sum n=\frac{7n(n+1)}{2}$ $[\because f(1)=7, given]$

63. For non-livial soln

D = 0

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$$\text{ie} \begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

$\Rightarrow 1(3bc-4bc)-2a(c-b)+a(4c-3b)=0$

$-bc+2ac-ab=0$

$\Rightarrow a, b, c$ in H.P

Now $\dfrac{\left(\frac{1}{a}\right)^{-3}+\left(\frac{1}{c}\right)^{-3}}{2} > \left(\dfrac{\frac{1}{a}+\frac{1}{c}}{2}\right)^{-3}$

$\Rightarrow \dfrac{a^3+c^3}{2} > b^3$

$\Rightarrow \dfrac{a^3+c^3}{b^3} > 2$

64. use projection formula

65. $p(x)=x^6-x^5-x^3-x^2-x$ &

$q(x)=x^4-x^3-x^2-x$

Hence $p(x)=x^2\left(x^4-x^3-x-1\right)-x$

$p(\alpha)=-\alpha,\ p(\beta)=-\beta$ etc.

$\therefore \sum p(\alpha)=-(\alpha+\beta+\gamma+\delta)=-1$

66. $\int x^x\left(\ln x(1+\ln x)+\dfrac{1}{x}\right)dx$

$\int \underbrace{x^x(1+\ln x)}_{II}\cdot \underbrace{\ln x}_{I}\,dx+\int x^{x-1}dx$

$=(\ln x)x^x-\int \dfrac{1}{x}\cdot x^x dx+\int x^{x-1}dx+c$

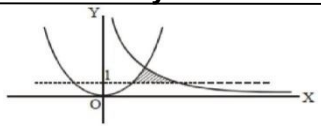
$=x^x\ln x$

67. For intersection, $\dfrac{8}{y}=\sqrt{y} \Rightarrow y=4$

Hence, required area $=\int_1^4\left(\dfrac{8}{y}-\sqrt{y}\right)dy$

$=\left[8\ln y-\dfrac{2}{3}y^{3/2}\right]_1^4=16\ln 2-\dfrac{14}{3}$

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68. $\left(x^4z^3 - z^3x^4\right)^{\top} = \left(z^{\top}\right)^3 \cdot \left(x^{\top}\right)^4 - \left(x^{\top}\right)^4 \cdot \left(z^{\top}\right)^3$

$= z^3x^4 - x^4z^3$

$= -\left(x^4z^3 - z^3x^4\right)$

$\left(x^{23} + x^{23}\right)^{\top} = \left(x^{\top}\right)^{23} + \left(y^{\top}\right)^{23}$

$= -x^{23} - y^{23}$

$= -\left(x^{23} + x^{23}\right)$

$\therefore$ C & D

69. $\tan\left(\cos^{-1}\dfrac{1}{5\sqrt{2}} - \sin^{-1}\dfrac{4}{\sqrt{17}}\right)$

$\tan\left(\tan^{-1}(7) - \tan^{-1}(4)\cdot\right)$

$\tan\left(\tan^{-1}\left(\dfrac{7-4}{1+28}\right)\right)$

$\tan\left(\tan^{-1}\left(\dfrac{3}{29}\right)\right) = \dfrac{3}{29}$

70. $\Rightarrow 2\cos\left(\dfrac{A-C}{2}\right) = 4\cdot\sin\dfrac{B}{2}$

$2\cos\left(\dfrac{A-C}{2}\right)\sin\left(\dfrac{A+C}{2}\right) = 4\sin\dfrac{B}{2}\cos\dfrac{B}{2}$

$\therefore \sin A + \sin C = 2\sin B$

$\therefore a + c = 2b$

71. $N = 2^3 \cdot 3^2$

$S = 1 + 2 + 2^2 - \cdots - 2^3 \cdot 3^2$

$R = 1 + \dfrac{1}{2} + \dfrac{1}{2^2} + \cdots - \dfrac{1}{2^3}\cdot\dfrac{1}{3^2}$

$R = \dfrac{3^2 \cdot 2^3 + 2^3 \cdot 3^2 .... + 1}{2^3 \cdot 3^2}$

$R = \dfrac{S}{N1}$

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$\therefore \frac{S}{R} = N = 2^3 \cdot 3^2 = 2(6)^2$

$\therefore a + b = 3$

72. $y = \left(\text{sen}^{-1}x\right)^2 + A\left(\cos^{-1}x\right) + B$

$$\frac{dy}{dx} = \frac{2\left(\sin^{-1}x\right)}{\sqrt{1-x^2}} - \frac{A}{\sqrt{1-x^2}}$$

$$\left(\sqrt{1-x^2}\right)\frac{dy}{dx} = 2\sin^{-1}x - A$$

$$\left(1-x^2\right)\left(\frac{dy}{dx}\right)^2 = \left(2\sin^{-1}x - A\right)^2$$

differentiate again

$$2\left(1-x^2\right)\frac{dy}{dx}\times\frac{d^2y}{dx^2} + (-2x)\left(\frac{dy}{dx}\right)^2 = 2\frac{\left(2\sin^{-1}x - A\right)}{\sqrt{1-x^2}}$$

$$2\left(1-x^2\right)\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} - 2x\left(\frac{dy}{dx}\right)^2 = 2\frac{dy}{dx}$$

$$\Rightarrow \left(1-x^2\right)\frac{d^2y}{dx^2} - \frac{dy}{dx} = 2$$

$\therefore P = 1, Q = 2$

73. tgt drawn at $A\left(x_1, y_1\right)$ meet x-axis at $2x_1$, hence area $= \frac{1}{2}|2x_1||y_1| = c^2$

$\therefore$ Rabio is 1

74. $$\lim_{x\to 0}\frac{\ln\left(x^4+x^2+1\right)}{\left(1-\cos^2x\right)}\cdot\cos x$$

$$\lim_{x\to 0}\frac{\ln\left(1+x^2+x^4\right)}{x^2+x^4}\cdot\frac{x^2\left(1+x^2\right)}{\sin^2x}\cdot\cos x$$

$= 1\cdot 1\cdot 1\cdot 1 = 1$

75. $f(gx) = x$

$$g'(x) = \frac{1}{f'g(x)}$$

$$g'(1) = \frac{1}{f'g(1)}$$

$$g'(1) = \frac{1}{f'(0)}$$

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